

(TR-104) PYTHON WITH AI/ML

Training Day 49 Report

Date: 19 March 2026

Topic Covered: Project Demonstration, Query Correction Improvements & Streamlit Concepts

The forty-ninth day of training focused on presenting the RAG-based AI Assistant project, improving query correction handling inside the pipeline, and learning advanced Streamlit concepts for building interactive AI applications.

- Delivered the project presentation explaining the complete RAG pipeline workflow, architecture, and implemented features.
- Demonstrated the live working of the AI Assistant project and showcased context-aware response generation.
- Identified that misspelled user queries were returning “No relevant context found” during retrieval.
- Explored the SymSpell library for spelling correction but observed incorrect corrections due to lack of domain awareness.
- Replaced SymSpell with an LLM-based query rewriter using the existing Ollama model already loaded in memory.
- Added domain-aware prompts with few-shot examples for accurate query correction before retrieval.
- Improved query understanding inside the RAG pipeline for better semantic retrieval performance.
- Practiced Streamlit sidebar components including selectbox and slider.
- Explored multiple display elements such as badge, caption, code block, divider, echo, text, help, and HTML rendering.
- Implemented column layouts with interactive button and radio selection handling.
- Practiced dynamic progress bar implementation using iterative updates.
- Created charts and maps using sample random datasets.
- Implemented multi-page navigation using sidebar-based routing.
- Learned Streamlit caching using `st.cache_data` and `st.cache_resource` for improving application performance.

- Understood session management concepts and how each user maintains separate session state in Streamlit applications.
- Strengthened understanding of frontend interaction handling for AI-powered applications.
- **GitHub Link:**
<https://github.com/JaspinderKaurWalia26/streamlit>
- **Blog Link:**
<https://medium.com/@jaspinderkaurwalia855/turning-python-into-apps-streamlit-practice-day-5bd3614612da>